Cambridge Assessment & Syllabus Assessment Standards

Advanced Level Biology Specification

BIOLOGY

9700 / AQA 3.2.2

Cell Division in Eukaryotic & Prokaryotic Cells

ASSESSMENT & MARKING SPECIFICATION GUIDE

Complete Syllabus Criteria & Indicative Standards

Published for Teachers and Candidates

This specification guide is published as an authoritative reference outlining the required learning outcomes, indicative marking points, and examiner guidance for the topic of Cell Division across Advanced Level specifications.

This document consists of 3 printed pages.

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Topic Section	Answer & Indicative Marking Criteria	Marks
1. The Cell Cycle & Interphase	**Interphase** accounts for >90% of the cell cycle ; **G1 phase:** cell growth, protein synthesis (enzymes, structural proteins), replication of organelles ; **S phase:** semi-conservative DNA replication; each chromosome now consists of two genetically identical sister chromatids joined at the centromere ; **G2 phase:** ATP synthesis, spindle tubulin protein synthesis, DNA repair and final growth ; regulated by cyclins and cyclin-dependent kinases (CDKs) at G1/S, G2/M, and spindle checkpoints ; *Examiner Note: Chromosome count remains 2n throughout interphase; DNA content doubles from 2C to 4C during S phase.*	**3**
2. Stages of Mitosis	**Prophase:** chromatin condenses into visible chromosomes; centrosomes migrate to opposite poles; nucleolus disappears; nuclear envelope disintegrates ; **Metaphase:** spindle microtubules attach to kinetochores at centromeres; chromosomes align singly along the spindle equator ; **Anaphase:** centromeres divide; spindle microtubules shorten / contract; sister chromatids pulled centromere-first to opposite poles ; **Telophase:** chromatids (now daughter chromosomes) reach poles and uncoil; nuclear envelope reforms around each group; spindle breaks down ; *Examiner Guidance: Reject references to 'homologous chromosomes separating' in mitosis (homologous chromosomes separate only in meiosis I).*	**4**
3. Cytokinesis	**Animal cells:** contractile ring of actin and myosin filaments forms under plasma membrane; creates a cleavage furrow that pinches cell into two ; **Plant cells:** rigid cell wall prevents cleavage; vesicles from Golgi apparatus align along equatorial plate and fuse forming cell plate ; vesicle contents synthesize middle lamella (pectin) and primary cellulose cell wall ; *Key Distinction: Cell plate formation progresses centrifugally (from center outward).*	**3**
4. Required Practical: Root Tip Squash & Mitotic Index	cut 1–2 mm from the apical meristem of growing plant root tip (e.g. onion / garlic) ; treat with warm dilute HCl / ethanoic alcohol to macerate tissue / hydrolyse middle lamella ; stain with acetic orcein / toluidine blue / Feulgen's stain to bind chromatin ; place on slide, add coverslip, press firmly down vertically with thumb (no sideways movement to avoid shearing cells) ; formula: *Mitotic Index = (number of cells with visible chromosomes / total number of cells observed) × 100* ; *Calculation Guidance: Count a large sample (>100 cells) across multiple fields of view to ensure statistical reliability.*	**5**
5. Cancer & Uncontrolled Cell Division	uncontrolled cell division results from mutations in genes controlling the cell cycle ; **proto-oncogenes** mutate into **oncogenes**, leading to over-stimulation of cell division ; mutation in **tumour suppressor genes** (e.g. p53) causes loss of cell cycle inhibition / failure of apoptosis ; cells divide repeatedly forming a tumour ; malignant tumours invade surrounding tissues and undergo metastasis via bloodstream or lymph vessels to form secondary tumours ; chemotherapy agents: vincristine (disrupts spindle microtubule formation in metaphase); fluorouracil (inhibits nucleotide synthesis in S phase) ;	**5**

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Topic Section	Answer & Indicative Marking Criteria	Marks
6. Binary Fission & Viral Replication	**Prokaryotic Binary Fission:** single circular DNA molecule replicates semi-conservatively; plasmids replicate; cell elongates; cytoplasm divides by inward growth of plasma membrane and peptidoglycan cell wall (septum) ; produces two genetically identical daughter bacterial cells ; **Viral Replication:** non-living / acellular; attachment proteins bind specific complementary receptors on host cell surface membrane ; injects viral nucleic acid into host cell; host cell metabolic machinery (ribosomes, enzymes, ATP) synthesizes viral capsid proteins and replicates viral genome ; assembly of new virions and release by cell lysis or budding ; *Comparison Rule: Viruses cannot undergo cell division / mitosis as they lack cellular organization, cytoplasm, and organelles.*	**4**

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